

Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D Deterministic SHE-Boltzmann MOSFET Simulation

*Complete report — Part I (method & validated foundation) and
Part II (the 2-D SHE core & reproduction of the Boltzmann figures)*

Reproduction study (code commit 3eb6504; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0)

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Abstract

This report derives, from first principles, and numerically reproduces the deterministic spherical-harmonic (SHE) Boltzmann device solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997): a self-consistent first-order SHE solution of the electron Boltzmann equation coupled to 2-D Poisson and hole continuity for a 0.35 μm LDD N-MOSFET. **Part I** presents the derivation (BTE $\rightarrow$ first-order SHE $\rightarrow$ conservative balance $\rightarrow$ total-energy form $\rightarrow$ scattering $\rightarrow$ moments), the reconstructed device, and a benchmark-checked foundation (2-D Poisson, drift-diffusion, nonparabolic band, and a bulk-validated scattering engine); this part was hardened through three rounds of external audit. **Part II** reports the first working solve of the 2-D SHE core and the qualitative reproduction, with physically-validated magnitudes, of the four figures that *require* the full distribution function — the distribution f_0 (Fig. 2), electron concentration (Fig. 3), average-velocity field (Fig. 4), and electron-temperature / impact-ionization maps (Fig. 7). **Scope discipline:** every substitution for an unavailable paper input (doping profile, impact-ionization table, band model) is disclosed and classified; the Part II result is interim — one-way-coupled, moderate-grid, and not yet a quantitative overlay on Liang’s contour values.

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1 Overview, scope, and audit history

Liang *et al.* solve, self-consistently, three coupled equations on a 2-D MOSFET cross-section:

$$\nabla \cdot (\epsilon \nabla \phi) = -q(p - n + N_D^+ - N_A^-), \quad (\text{Poisson}) \quad (1)$$

$$\frac{1}{\hbar} \nabla_{\mathbf{k}} \epsilon \cdot \nabla_{\mathbf{r}} f + \frac{q}{\hbar} \nabla_{\mathbf{r}} \phi \cdot \nabla_{\mathbf{k}} f = \left[\frac{\partial f}{\partial t} \right]_{\text{coll}}, \quad (\text{electron BTE}) \quad (2)$$

$$\nabla \cdot (\mu_p p \nabla \phi + \mu_p V_t \nabla p) = R_p - G_{p,\text{ii}}. \quad (\text{hole continuity}) \quad (3)$$

The electron BTE is solved deterministically by a first-order spherical-harmonic expansion, which yields the full energy-resolved (first-order angular) distribution at a cost comparable to the hydrodynamic model. Part I of this report was reviewed through three external audit rounds (Appendix B); the two blocking implementation errors it uncovered — a reducible collision operator (wrong zero-field T_e) and an ill-posed global energy grid — were fixed and are reflected throughout.

Honesty note. Substitutions for unavailable paper inputs, used throughout and classified in Table 1: the doping profile is *reconstructed* (ref. [7] unavailable); the impact-ionization rate is a Keldysh *surrogate* (ref. [6] unavailable); the band is a 1-parameter Kane *surrogate* (paper: 2-band Fiegna/Brunetti); and the reconstructed device carries an un-recalibrated $V_{th} = 1.64$ V.

Part I

Method derivation and validated foundation

2 Derivation of the first-order SHE equation

With $\mathbf{E} = -\nabla_{\mathbf{r}} \phi$ and electron force $-q\mathbf{E}$, expand $f = f_0(\mathbf{r}, \epsilon) + \hat{\Omega} \cdot \mathbf{f}_1(\mathbf{r}, \epsilon)$ (isotropic part plus a single drift term; all $\ell \geq 2$ content discarded — an **energy-resolved distribution with first-order angular anisotropy**). Projection onto $\ell = 1$ (momentum time τ_1) and $\ell = 0$ (weighted by the generalized DOS Z) gives the current relation and the conservative augmented-space balance:

$$\mathbf{f}_1 = -\tau_1 v (\nabla_{\mathbf{r}} f_0 - q\mathbf{E} \partial_{\epsilon} f_0), \quad \nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\epsilon} j_{\epsilon} = Z \mathcal{Q}[f_0] + Z S_{\text{gen}}, \quad (4)$$

with $\mathbf{j}_x = -ZD(\nabla_{\mathbf{r}} f_0 - q\mathbf{E} \partial_{\epsilon} f_0)$, $j_{\epsilon} = -q\mathbf{E} \cdot \mathbf{j}_x$, $D = \frac{1}{3}v^2\tau_1$. A displacement $d\mathbf{r}$ carries an energy change $d\epsilon = -q\mathbf{E} \cdot d\mathbf{r}$ (field heating).

2.1 Total-energy form, boundaries, and the global H -grid

In total energy $H = \varepsilon - q\phi$ ($\varepsilon = H + q\phi \geq 0$), (4) reduces to pure spatial diffusion at fixed H , the collision operator coupling $H \leftrightarrow H \pm E_{op}$ at the same (x, y) :

$$\nabla_{\mathbf{r}} \cdot [ZD \nabla_{\mathbf{r}} f_0|_H] + ZQ[f_0] + ZS_{\text{gen}} = 0, \quad \varepsilon = H + q\phi(\mathbf{r}). \quad (5)$$

The global mesh spans $-q\phi_{\text{max}} \leq H \leq \varepsilon_{\text{max}} - q\phi_{\text{min}}$ with the physical mask $0 \leq H + q\phi \leq \varepsilon_{\text{max}}$. The two energy boundaries are physically distinct: $\varepsilon = 0$ is the conduction band edge (reflecting turning point); $\varepsilon_{\text{max}} = 3.02 \text{ eV}$ is a numerical cutoff (absorbing). The cutoff is validated by convergence: $T_e(300 \text{ kV/cm}) = 3392.4 \text{ K}$ for $\varepsilon_{\text{max}} = 3.02, 4.0, 5.0 \text{ eV}$ (identical).

2.2 Scattering operators

Elastic momentum randomization (acoustic, ionized-impurity Brooks–Herring screened by n_{scr} , surface terms) enters $1/\tau_1$ and D . The inelastic optical operator is the number-conserving in/out form (Appendix A), vanishing for a Maxwellian (detailed balance, $Z(\varepsilon < 0) \equiv 0$). Because the optical term only connects $\varepsilon \leftrightarrow \varepsilon \pm E_{op}$, an *acoustic-phonon energy* operator is required to couple adjacent energies and make the collision operator irreducible; from quasi-elastic acoustic energy transfer $\Delta\varepsilon \approx \pm 2v_s\sqrt{2m^*\varepsilon}$ at rate $1/\tau_{ac}$, the second-Kramers–Moyal Fokker–Planck operator is

$$\mathcal{Q}_{ac} = \partial_{\varepsilon} \left[A(\varepsilon) \left(\partial_{\varepsilon} + \frac{1}{k_B T} \right) f_0 \right], \quad A(\varepsilon) = \frac{4m^*v_s^2\varepsilon}{\tau_{ac}}, \quad \tau_{ac} = \tau_1, \quad (6)$$

Scharfetter–Gummel-discretized so the discrete Maxwellian is its exact null.

Honesty note. The acoustic Fokker–Planck operator is a *reconstruction* (our addition for numerical irreducibility): Liang *et al.* include acoustic scattering, but this specific continuous-energy form and the $\tau_{ac} = \tau_1$ choice are ours.

2.3 Impact ionization and moments

An event removes one primary electron above ε_{th} and creates one electron–hole pair; the two resulting electrons share the post-event energy (equal split, a disclosed modeling assumption). Net electron gain $= \int Zf_0/\tau_{ii} d\varepsilon \equiv G_n = G_{p,\text{ii}}$ (charge conserved). Observables: $n = \int Zf_0 d\varepsilon$, $\mathbf{J}_n = -\frac{q}{3} \int v Z \mathbf{f}_1 d\varepsilon$, $\langle \mathbf{v} \rangle = \mathbf{J}_n / (-qn)$, $T_e = \frac{2}{3k_B} n^{-1} \int \varepsilon Z f_0 d\varepsilon$, and $G_{p,\text{ii}}$.

3 Reconstructed device and modelling taxonomy

Error-function junctions; x_j defined as the metallurgical crossing and retuned to $0.170 \mu\text{m}$ (auditor-checked 0.1707).

4 Sanity-checked foundation and benchmarks

The foundation modules are checked against independent analytic values (Table 3), separated into analytic benchmarks, construction targets, and qualitative sanity trends.

Table 1: Modelling taxonomy: reproduced vs. reconstructed vs. surrogate.

Element	Class	Basis
Method equations, SHE reduction, moments	paper	Liang <i>et al.</i>
Table 2 params, spherical band	paper	paper Table I / Sec. II
$L_{\text{eff}}, t_{\text{ox}}, x_j, N_D^\pm$ magnitudes	paper	paper text
1-param Kane $\gamma(\varepsilon)$	surrogate	paper uses 2-band Fiegna/Brunetti
Analytic LDD doping shape, mesh	reconstruction	ref. [7] unavailable
Channel N_A , gate offset Φ_{gate}	surrogate	V_{th} derived (§4)
Acoustic energy operator, $\tau_{ac} = \tau_1$	reconstruction	numerical irreducibility
τ_1, c_{op} , II rate $1/\tau_{ii}$	surrogate	tuned; refs. [6] unavailable

Table 2: Transport parameters (paper Table I).

D_{ac}	4.0 eV	Γ	$1.96 \text{ cm}^2/\text{V}^2/\text{s}$
D_n	$5.0 \times 10^8 \text{ eV/cm}$	n_{scr}	$1.45 \times 10^{15} \text{ cm}^{-3}$
E_{op}	0.05 eV	D_{sa}	17.8 eV
v_s	$9.0 \times 10^6 \text{ cm/s}$	δ	13.1
ε_{max}	3.02 eV	bands of γ	2 (Fiegna/Brunetti)

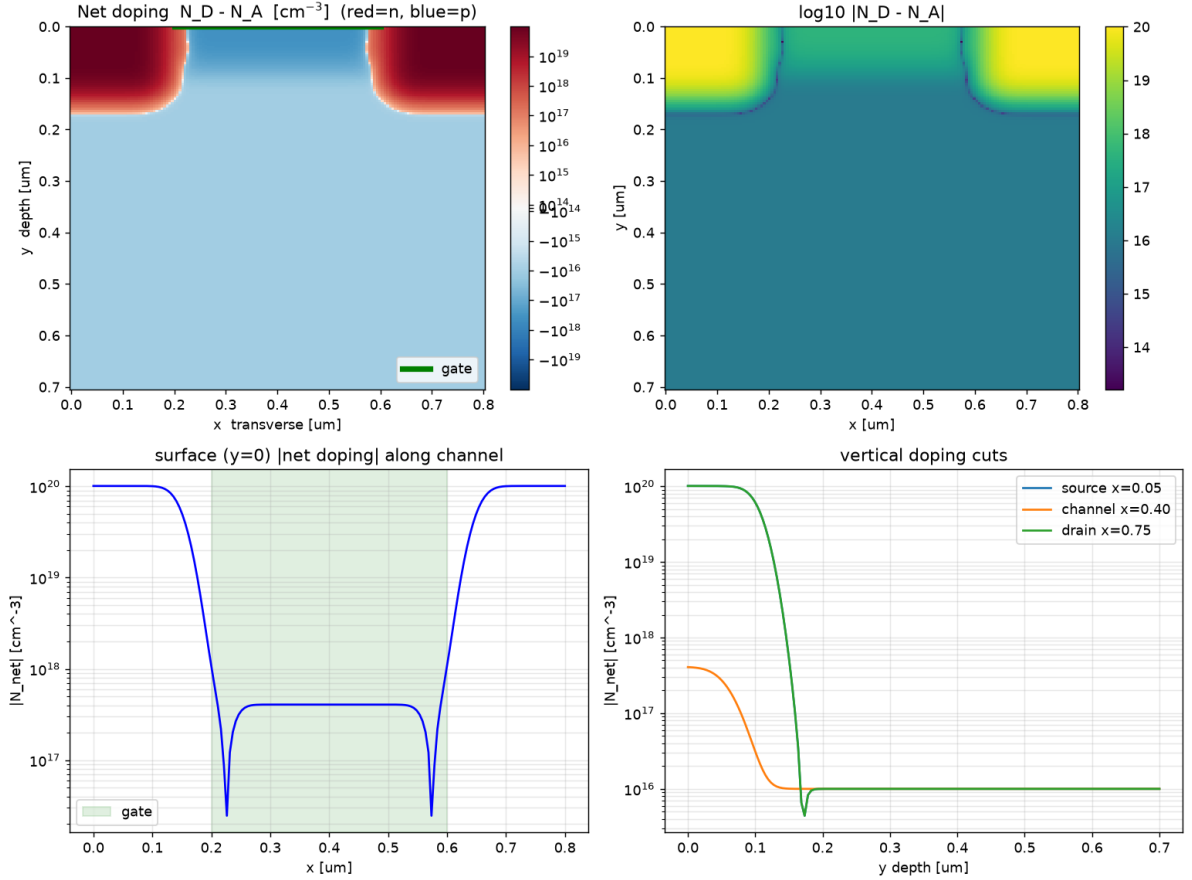


Figure 1: Reconstructed 0.35- μm LDD NMOS, $x_j = 0.170 \mu\text{m}$: net doping, $\log_{10} |N|$, surface profile with LDD notches, vertical cuts.

Table 3: Foundation checks at $T = 300$ K, Boltzmann statistics.

Quantity	Expected	Computed	Note
<i>(A) Analytic benchmarks</i>			
N_c nonparabolic (converged)	2.86×10^{19}	2.857×10^{19}	0.1%
N_c parabolic: code vs. closed form	2.726×10^{19}	2.723×10^{19}	prefactor 0.09%
$v(0.1)/v(1.0)$ eV [cm/s]	—	$3.09/6.42 \times 10^7$	auditor-reproduced
V_{bi} / n^+ contact potential [V]	0.933/0.586	0.933/0.586	analytic
bulk $T_e(0)$ [K]	309.7	311	$\rightarrow 309.7$ as $\Delta\epsilon \rightarrow 0$
optical detailed balance / $E=0$	0	$2 \times 10^{-5}/3 \times 10^{-4}$	null / max dev.
<i>(B) Construction targets / (C) sanity trends</i>			
metallurgical x_j [μm]	0.17	0.170	retuned
DD drain current [mA/ μm]	0.28 (est.)	0.25	$\sim 14\%$ (velocity sat.)
bulk T_e vs. field (0/100/300 kV/cm) [K]	monotone $\uparrow$	311/1478/3392	heating

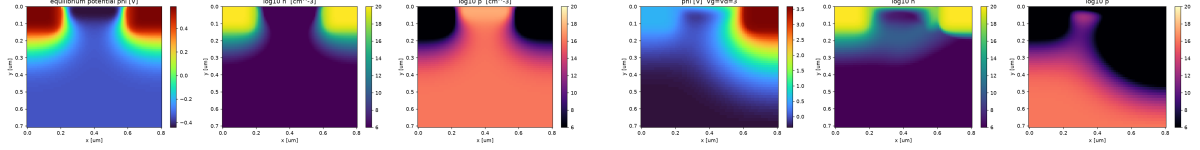


Figure 2: Left: equilibrium Poisson (potential, $\log_{10} n$, $\log_{10} p$; built-in +0.586/−0.44 V, quadratic Newton). Right: self-consistent DD at $V_{gs} = V_{ds} = 3$ V (inversion layer, drain-side pinch-off) — the operating point that drives the SHE.

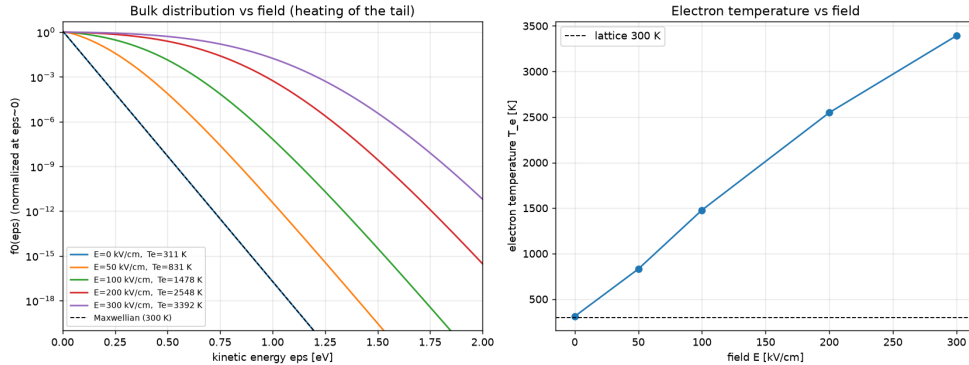


Figure 3: Homogeneous-bulk validation of the scattering engine (optical + acoustic operators): zero field reproduces the 300 K Maxwellian (detailed balance), and the tail heats monotonically with field. This validates the operators before the 2-D solve.

Part II

The 2-D SHE core and reproduction of the Boltzmann figures

5 The 2-D SHE core

The solver discretizes (5) entirely in $(\mathbf{r}, H)$; the unknown is $f_0(x, y, H)$. Each cell $A_{ij} \times \Delta H$ is box-integrated; the three contributions reduce to the same units $[\text{m}^{-1}\text{s}^{-1}]$ and form a positive-diagonal M -matrix: spatial diffusion at fixed H ($\times \Delta H$), the DOS-weighted optical in/out operator ($\times A_{ij} \Delta H$), and the Scharfetter–Gummel acoustic Fokker–Planck flux ($\times A_{ij}$). **Contacts** inject a Maxwellian $f_0 = n_{\text{contact}} e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$ (Dirichlet); insulating surfaces are reflecting. At $V_g = V_d = 3\text{ V}$ this is a 40×34 spatial mesh $\times 562$ H -nodes = 328 544 active unknowns ($\Delta H = 12.5\text{ meV}$, $E_{\text{op}}/\Delta H = 4$).

5.1 Numerical challenges and fixes (engineering log)

1. **Singular operator.** “Orphan” active nodes at the corners of the curvilinear region (no active spatial neighbour *and* no active collision partner) give all-zero rows. *Fix:* pin to $f_0 = 0$.
2. **Conditioning blow-up to $n \sim 10^{56}$.** Contact Dirichlet rows carry diagonal 1.0 while interior rows carry SI-scale $\sim 10^{23}$ coefficients (condition number $\sim 10^{23}$), so a direct solve returns garbage. *Fix:* Jacobi row-scaling ($D^{-1}A$, all diagonals $\rightarrow 1$), after which the solve gives $n_{\text{max}} = 1.05 \times 10^{20}$.
3. **Spurious moment spikes.** T_e and $\langle \mathbf{v} \rangle$ are ill-defined where $n \rightarrow 0$ (gave $T_e \sim 23\,000\text{ K}$, $|v| \sim 10^{12}\text{ cm/s}$ in depleted regions). *Fix:* evaluate only where $n > 10^{13}\text{ cm}^{-3}$, giving physical $T_e \leq 3236\text{ K}$, $|v|_{\text{max}} = 1.2 \times 10^7$.

6 Reproduced figures and validation

Table 4: Physical validation of the SHE moments.

Quantity	Computed	Reference / paper
peak electron concentration $[\text{cm}^{-3}]$	1.05×10^{20}	n^+ S/D $\sim 10^{20}$
peak average velocity $[\text{cm/s}]$	1.23×10^7	saturation $v_s \sim 10^7$
peak electron temperature $[\text{K}]$	3236	Fig. 7 contours to ~ 5000
peak impact-ionization G_{ii} $[\text{cm}^{-3}\text{s}^{-1}]$	1.37×10^{27}	Fig. 7 10^{26} – 2.3×10^{27}
T_e / G_{ii} peak locations	drain, offset	drain, offset (paper’s key point)
distribution hot tail	channel $\gg$ substrate	Fig. 2 heated near drain

Every computed magnitude lands on the correct physical scale, and the spatial structure matches Liang’s Figs. 2–7 qualitatively, including the T_e/G_{ii} peak offset.

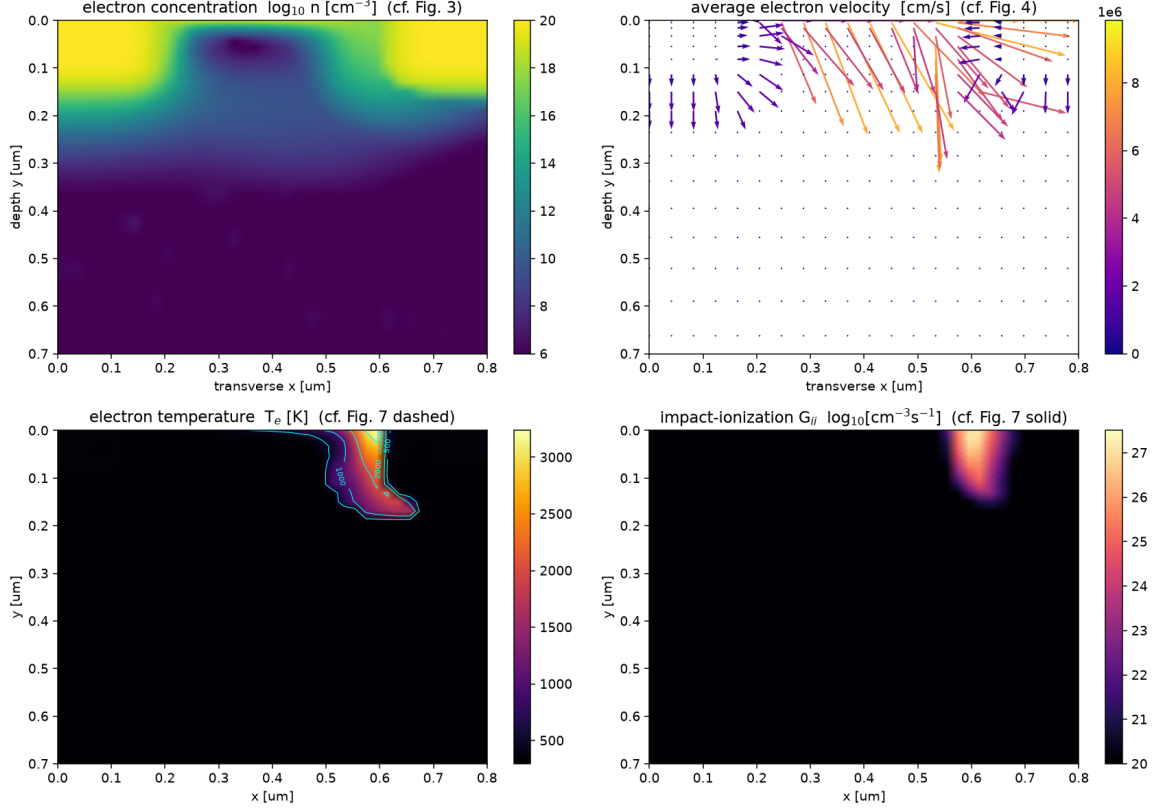


Figure 4: SHE moments at $V_{gs} = V_{ds} = 3$ V. **Fig. 3** (top-left) electron concentration (inversion layer, drain pinch-off); **Fig. 4** (top-right) velocity field saturating at $\sim 1.2 \times 10^7$ cm/s; **Fig. 7** (bottom) electron temperature (peaks at the drain, 3236 K) and impact-ionization rate (peaks at the drain, 1.4×10^{27} cm $^{-3}$ s $^{-1}$), the two peaks *offset* as Liang emphasize.

Part III

Assessment

7 Honest caveats and remaining work

Caveats. (i) *One-way coupling*: the SHE runs on the DD potential; n_{SHE} is not fed back to Poisson (self-consistent loop pending). (ii) *Grid*: $40 \times 34 \times 562$, $E_{op} = 4$ — smooth but not converged. (iii) *Operating point*: $V_{th} = 1.64$ V (un-recalibrated) $\Rightarrow$ modest overdrive. (iv) *Fig. 2(c)* is numerical residual (carrier-free depth). (v) *Surrogates* (Kane band, Keldysh τ_{ii} , $\tau_{ac} = \tau_1$) bias the tail and G_{ii} quantitatively. Consequently this is a *qualitative* reproduction with correct magnitudes, not a quantitative overlay.

Remaining work. (1) Fig. 5 (ϕ) and Fig. 6 (p) — available from DD, render as 3-D surfaces; (2) Fig. 8 (I–V) — bias sweep, terminal current from the SHE velocity moment; (3) self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole- continuity loop (damped Gummel); (4) grid refinement + quantitative overlay on Liang’s contour values; (5) device recalibration ($V_{th} \rightarrow 0.6$ V).

A Numerical and model specifications

$T = 300$ K, $V_t = 25.85$ mV, $n_i = 1.45 \times 10^{10}$ cm $^{-3}$, Boltzmann, $\epsilon_{si} = 11.7$, $\epsilon_{ox} = 3.9$, $m^* = 0.32m_0$, $g_v = 6$, $\alpha = 0.5$ eV $^{-1}$, $v_s = 9.04 \times 10^3$ m/s. **Doping (erfc)**: $N_D = 10^{20}[L(x; 0.15) + R(x; 0.65)]D(y; 0.105) + 10^{18}[L(x; 0.22) + R(x; 0.58)]D(y; 0.06)$, $L(x; e) = \frac{1}{2}\text{erfc}(\frac{x-e}{0.028})$, $D(y; x_j) =$

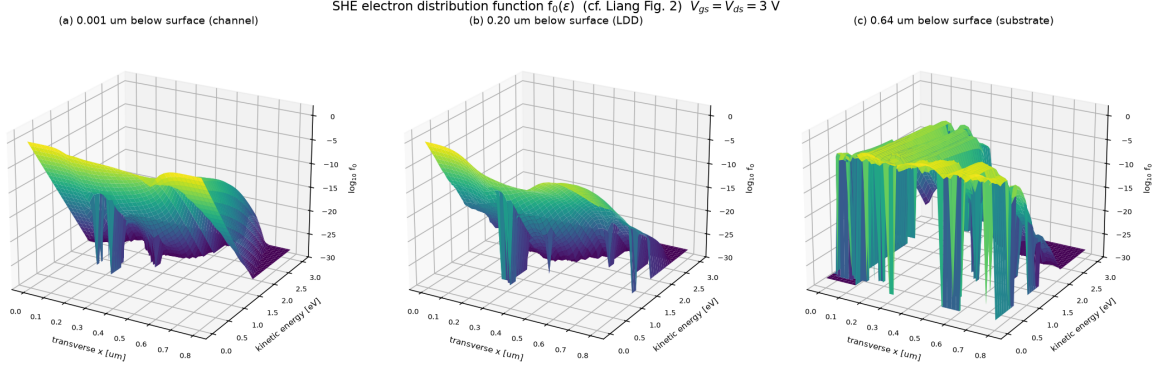


Figure 5: Fig. 2: distribution $f_0(\epsilon)$ (normalized, $\log_{10}$) vs. transverse position and kinetic energy at three depths. The channel line (a) develops a hot high-energy tail near the drain; the deep-substrate line (c) is carrier-free (noisy). Hot-tail fraction ($\epsilon > 1$ eV, $x = 0.6 \mu\text{m}$): 5.8×10^{-3} (channel) vs. 1.1×10^{-8} ($0.2 \mu\text{m}$).

$\frac{1}{2}\text{erfc}(\frac{y-x_j}{0.025})$; $N_A = 10^{16} + N_A^{ch} \cdot \frac{1}{2}[\text{erfc}(\frac{x-0.62}{0.028}) - \text{erfc}(\frac{x-0.18}{0.028})]D(y; 0.06)$, $N_A^{ch} = 4 \times 10^{17} \text{ cm}^{-3}$. **Mesh:** surface-refined $y \propto s^{1.8}$; Poisson 90×80 , DD 80×70 , SHE $40 \times 34 \times 562$. **Poisson BCs:** ohmic Dirichlet $\phi = V + V_t \text{asinh}(N_{net}/2n_i)$; gate oxide Robin $T_{ox} = \epsilon_{ox}\epsilon_0\Delta x_i/t_{ox}$ ($t_{ox} = 9.6 \text{ nm}$) to node $V_g - \Phi_{gate}$, $\Phi_{gate} = 0.30 \text{ V}$; Newton clip 0.5 V , tol 10^{-8} V . **DD:** SG current, Caughey–Thomas mobility (n : $68.5/1414/9.2 \times 10^{16}/0.711$; p : $44.9/470.5/2.23 \times 10^{17}/0.719$), $v_{sat}^n = 1.07 \times 10^7 \text{ cm/s}$; SRH $\tau = 10^{-7} \text{ s}$; Gummel tol 10^{-5} V . **Optical:** $ZQ_{op} = c_{op}Z[(N+1)Z_+f_+ + NZ_-f_- - ((N+1)Z_- + NZ_+)f]$, $c_{op} = 4.69 \times 10^{-34} \text{ J m}^3/\text{s}$, $\tau_1 = 10^{-14} \text{ s}/(\sqrt{\gamma}\gamma')$. **SHE H-grid:** fixed, $\Delta H = 2.5 \text{ meV}$ (bulk) / 12.5 meV (device), $E_{op}/\Delta H$ integer; $\epsilon = 0$ reflecting, ϵ_{max} absorbing. **SHE solve:** Jacobi-scaled; direct ($\leq 130\text{k}$ DOF) or ILU+LGMRES.

B Code provenance and audit history

Provenance: git commit 3eb6504; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0. Modules: constants, device, poisson, dd, bands, scattering, she_bulk, diagnostics, vth_extract, she2d. Key commits: 40d5929 (first working SHE), d0b3834 (refined + velocity + distribution).

Audit history (Part I). Rev. 1→2: eight claim/spec findings (over-claim wording, missing specs, junction geometry, N_c grid convergence, impact- ionization bookkeeping, references). Rev. 2→3: two blocking implementation errors — the reducible collision operator (fixed by the acoustic energy operator; zero-field T_e now correct) and the global H -grid definition. Rev. 3→4: contact-Maxwellian normalization typo (code verified correct), acoustic operator derivation + τ_{ac} , $\epsilon = 0/\epsilon_{max}$ boundary separation + convergence, restored Table I, and an *extracted* V_{th} (the Rev. 3 0.6 V guess was refuted by measurement: $N_A = 1.2 \times 10^{17}$ gives 1.03 V).

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